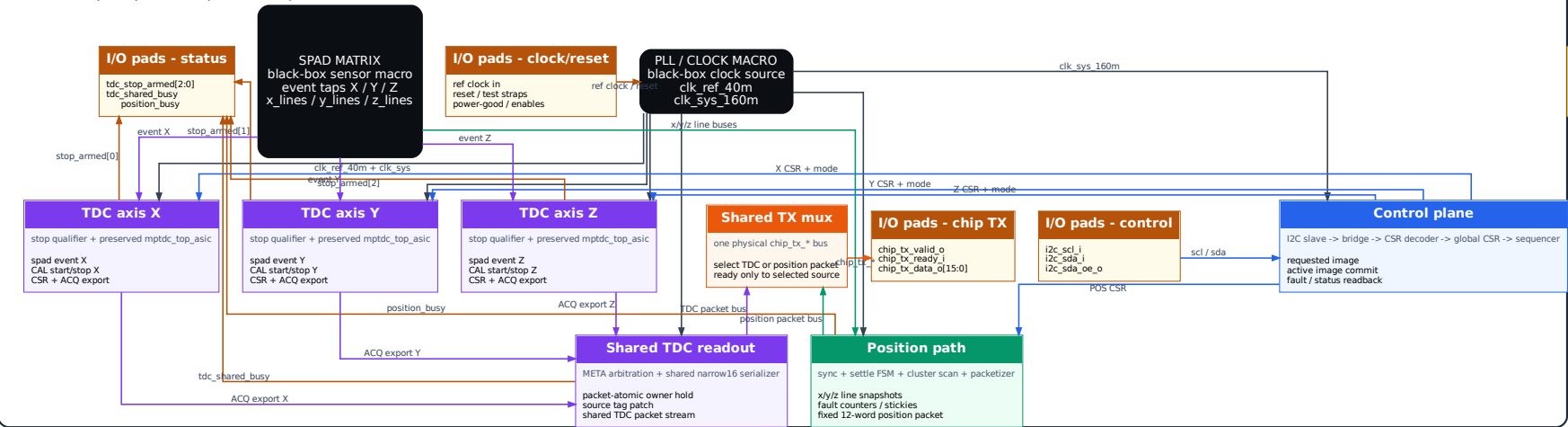


SPADMIC top conceptual floorplan schematic

Professional chip-level page with grouped I/O pads, black-box SPAD matrix, black-box PLL, and the active shared-bus architecture.
TOP/rtl/spadmic_top_v1.sv + TOP/docs/01_ACTIVE_ARCHITECTURE.md

SPADMIC top chip - conceptual floorplan schematic



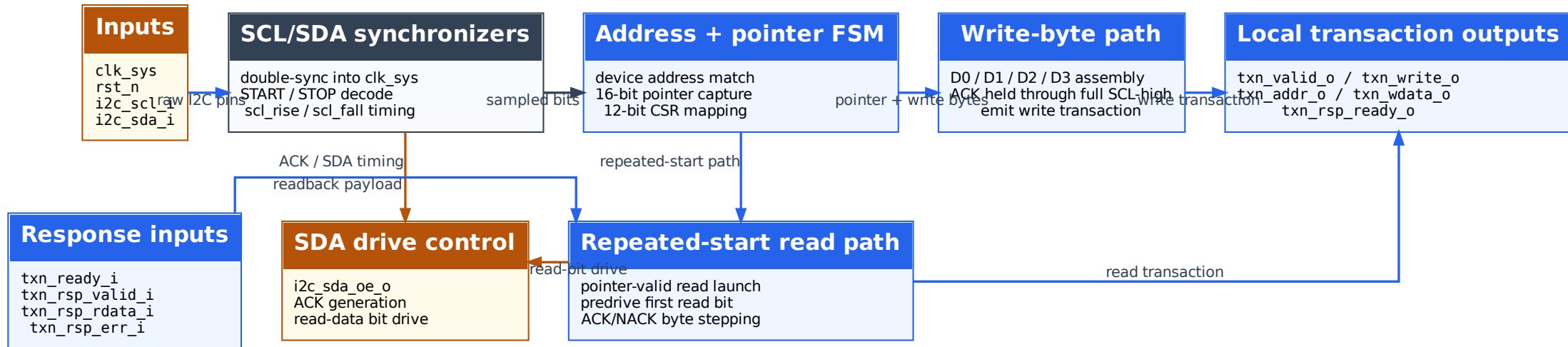
Technical note

The SPAD matrix and PLL are shown as black-box macros for chip-level communication clarity. The active RTL still treats clocks and matrix outputs as top-level interfaces. Three preserved TDC kernels stay local until acquisition-record export.

spadm_i2c_slave

Schematic page for the synchronized I2C front end and its pointer-based command/readback behavior.

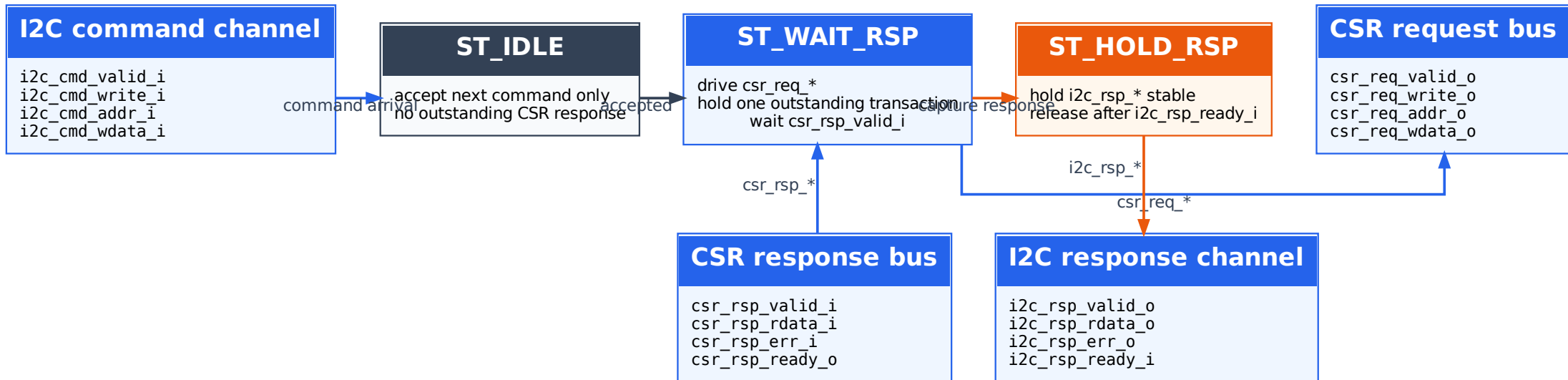
I2C/rtl/spadm_i2c_slave.sv



spadmic_i2c_csr_bridge

Single-outstanding bridge between the I2C transaction channel and the shared local CSR bus.

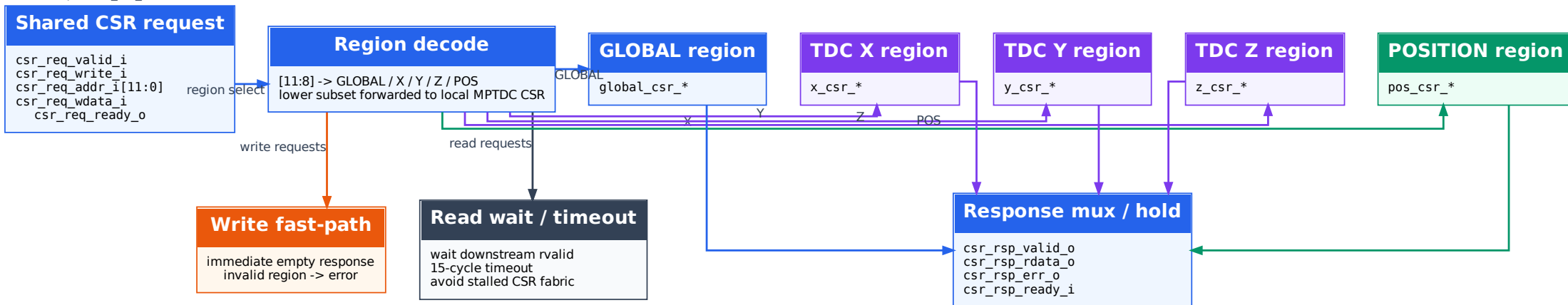
I2C/rtl/spadmic_i2c_csr_bridge.sv



spadmic_csr_decoder

Shared 12-bit CSR routing page with timeout-protected read behavior and explicit invalid-region handling.

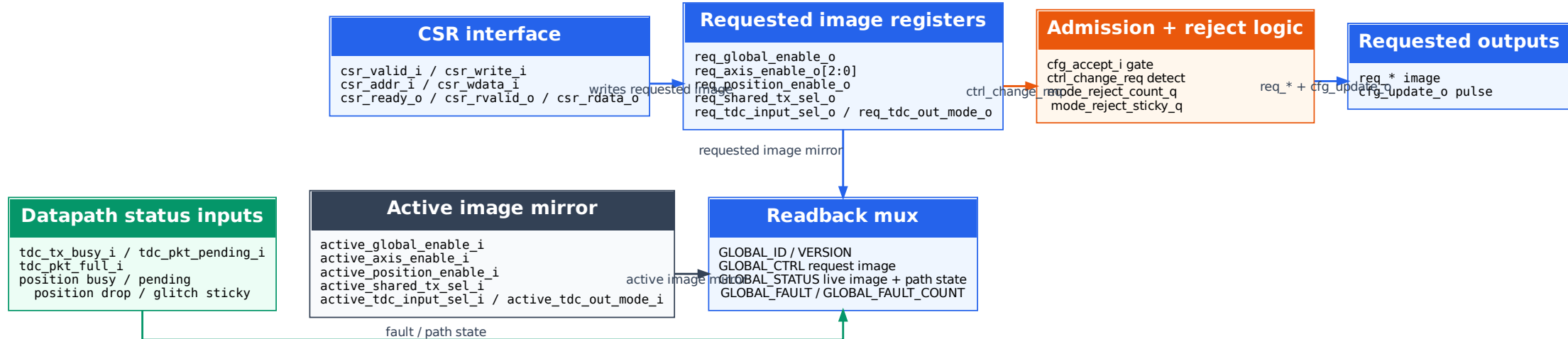
TOP/rtl/spadmic_csr_decoder.sv



spadmic_global_csr

Requested-control bank with active-state mirror, path-idle visibility, and counted reject reporting.

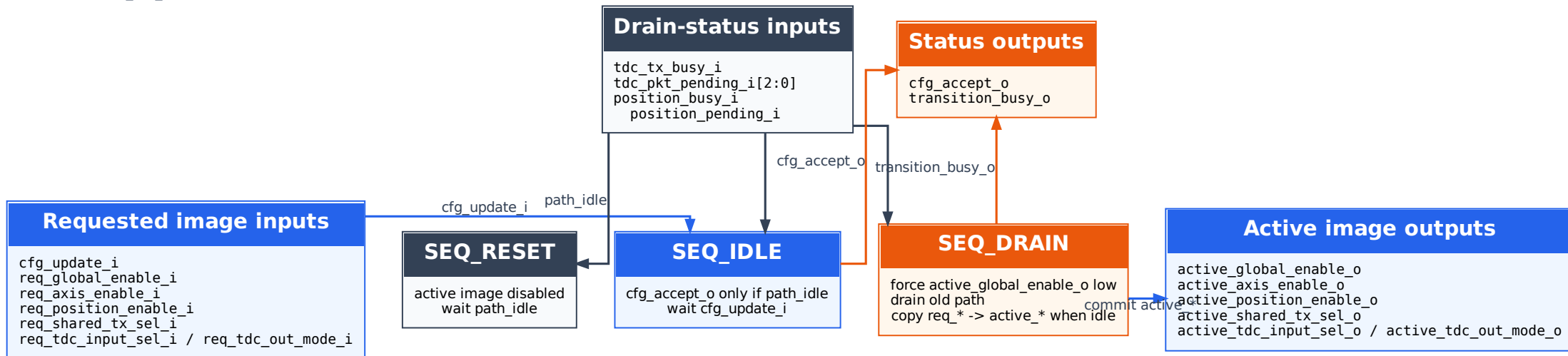
TOP/rtl/spadmic_global_csr.sv



spadmic_top_sequencer

Drain-aware requested-to-active handoff that keeps mode/source changes packet-safe.

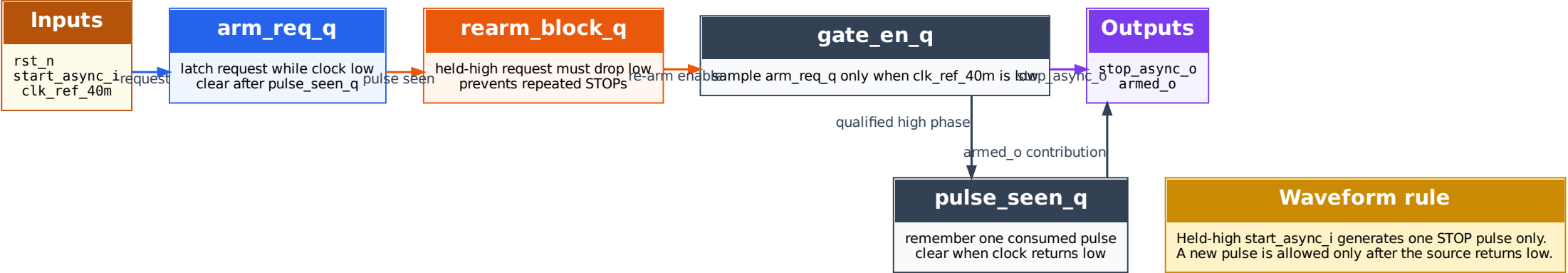
TOP/rtl/spadmic_top_sequencer.sv



spadmic_ref_stop_qualifier

One-shot STOP qualifier with explicit held-high re-arm blocking.

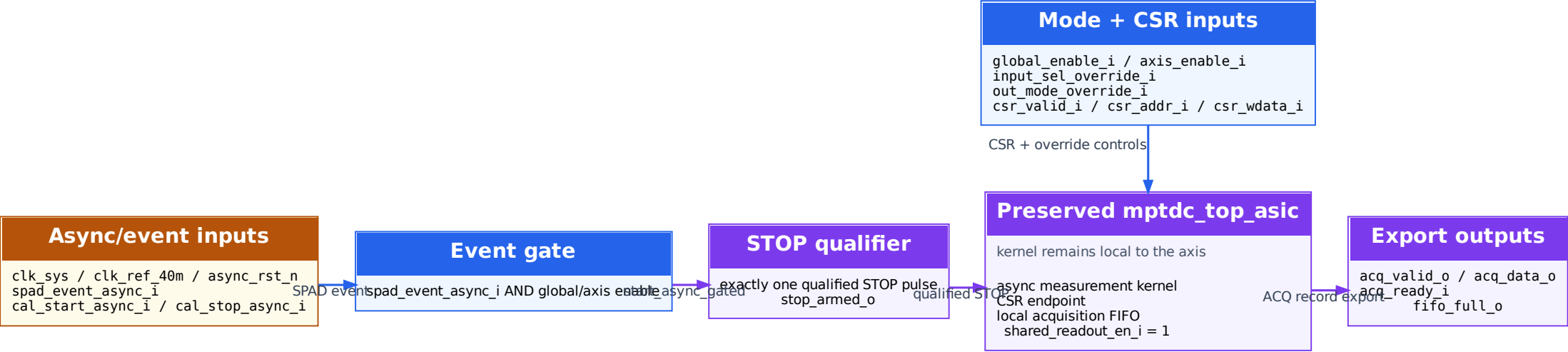
TOP/rtl/spadmic_ref_stop_qualifier.sv



spadmic_tdc_axis_wrapper

Per-axis wrapper around the qualifier and one preserved MPTDC kernel, exporting acquisition records to the top-level shared readout.

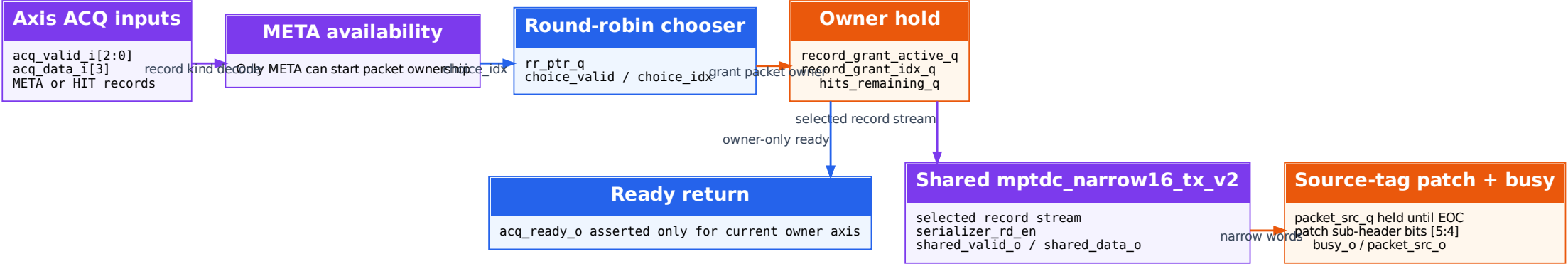
TOP/rtl/spadmic_tdc_axis_wrapper.sv



spadmic_tdc_shared_readout

META-first shared serializer front end with packet-atomic ownership and stable source tagging.

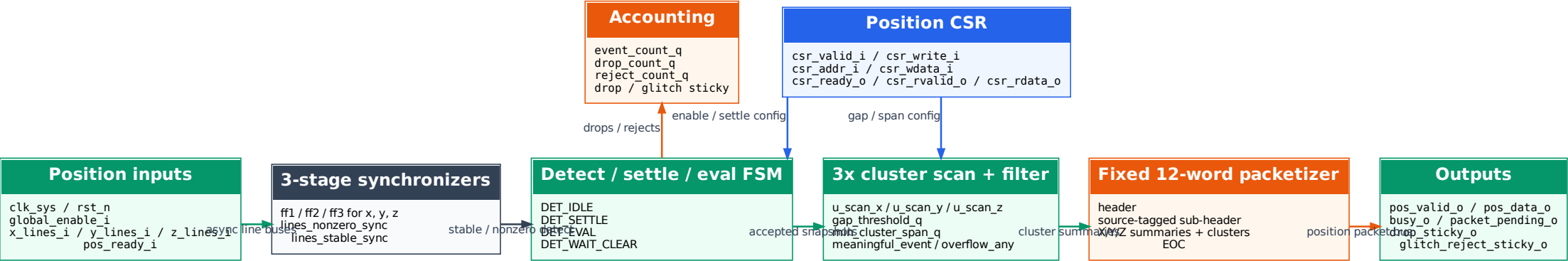
TOP/rtl/spadmic_tdc_shared_readout.sv



spadmic_position_block

Async-qualified position detector with explicit settle filtering, cluster extraction, and fixed packet output.

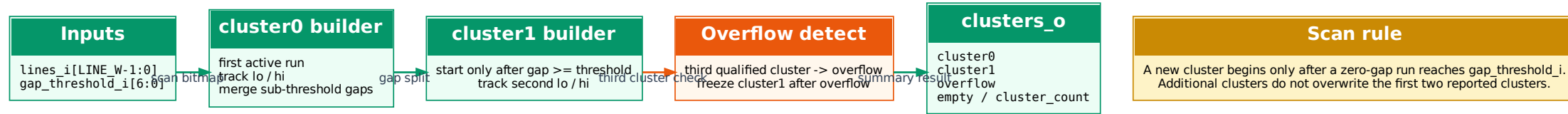
TOP/rtl/spadmic_position_block.sv



spadmic_axis_cluster_scan

Two-cluster bitmap scanner with threshold-based split and overflow retention of the first two clusters.

TOP/rtl/spadmic_axis_cluster_scan.sv



spadmic_shared_tx_mux

Final one-of-two packet mux onto the chip output interface.

TOP/rtl/spadmic_shared_tx_mux.sv

